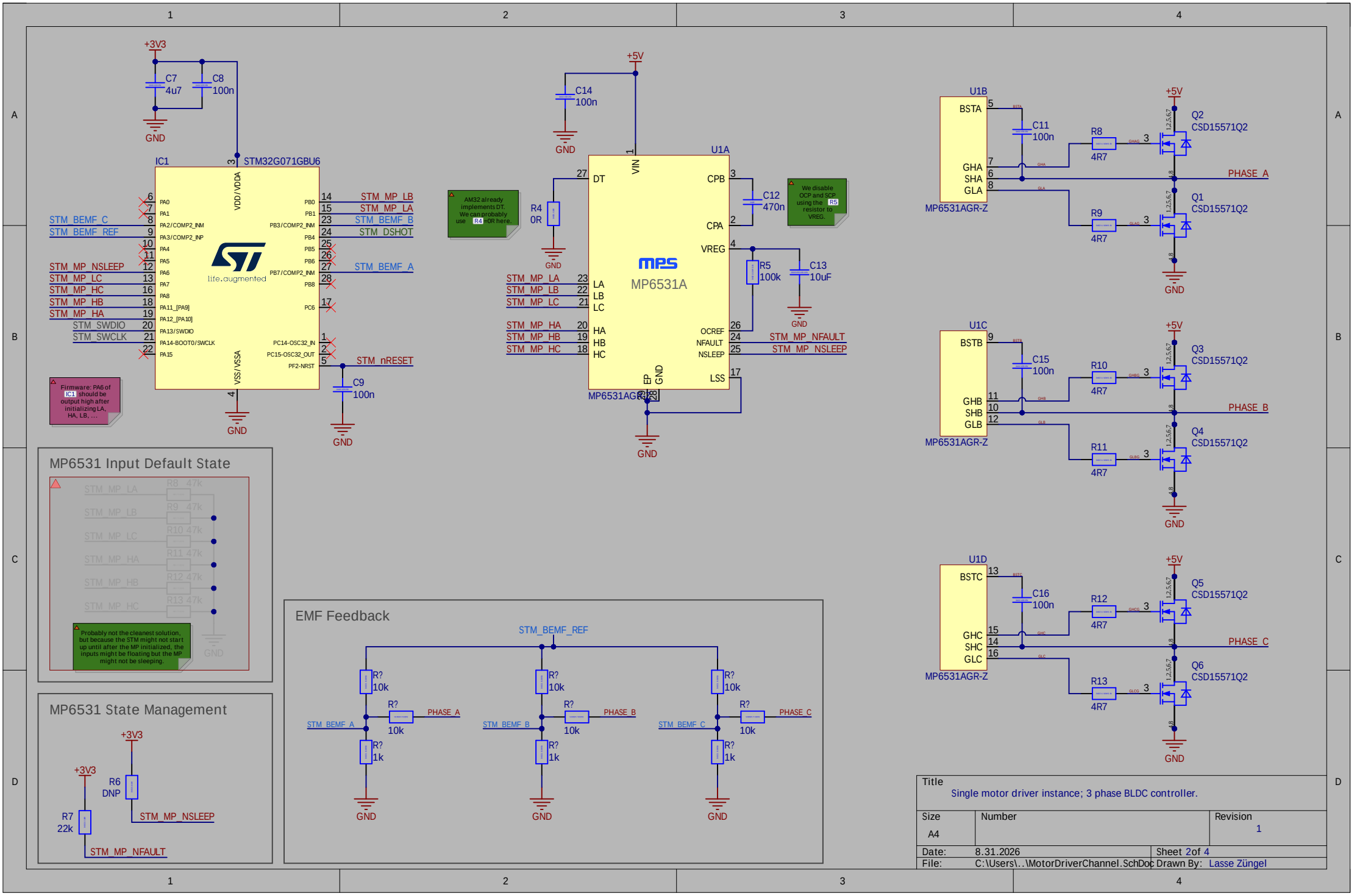

+3V3
C7
4u7
C8
100n
GND
IC1
STM32G071GBU6
VDD/VDDA
VSS/VSSA
PA0
PA1
PA2/COMP2_INM
PA3/COMP2_INP
PA4
PA5
PA6
PA7
PA8
PA11_[PA9]
PA12_[PA10]
PA13/SWDIO
PA14-BOOT0/SWCLK
PA15
PB0
PB1
PB3/COMP2_INM
PB4
PB5
PB6
PB7/COMP2_INM
PB8
PC6
PC14-OSC32_IN
PC15-OSC32_OUT
PF2-NRST
STM_BEMF_C
STM_BEMF_REF
STM_MP_NSLEEP
STM_MP_LC
STM_MP_HC
STM_MP_HB
STM_MP_HA
STM_SWDIO
STM_SWCLK
STM_MP_LB
STM_MP_LA
STM_BEMF_B
STM_DSHOT
STM_BEMF_A
STM_nRESET
C9
100n
Firmware: PA6 of IC1 should be output high after initializing LA, HA, LB, ...
+5V
C14
100n
U1A
MP6531A
VIN
DT
CPB
CPA
VREG
LA
LB
LC
HA
HB
HC
OCREF
NFAULT
NSLEEP
LSS
EP
GND
C12
470n
C13
10uF
R5
100k
R4
0R
AM32 already implements DT. We can probably use R4 =0R here.
We disable OCP and SCP using the R5 resistor to VREG.
STM_MP_LA
STM_MP_LB
STM_MP_LC
STM_MP_HA
STM_MP_HB
STM_MP_HC
STM_MP_NFAULT
STM_MP_NSLEEP
U1B
U1C
U1D
MP6531AGR-Z
BSTA
GHA
SHA
GLA
BSTB
GHB
SHB
GLB
BSTC
GHC
SHC
GLC
C11
100n
C15
100n
C16
100n
R8
4R7
R9
4R7
R10
4R7
R11
4R7
R12
4R7
R13
4R7
Q1
CSD15571Q2
Q2
CSD15571Q2
Q3
CSD15571Q2
Q4
CSD15571Q2
Q5
CSD15571Q2
Q6
CSD15571Q2
PHASE_A
PHASE_B
PHASE_C
MP6531 Input Default State
R8 47k
R9 47k
R10 47k
R11 47k
R12 47k
R13 47k
Probably not the cleanest solution, but because the STM might not start up until after the MP initialized, the inputs might be floating but the MP might not be sleeping.
MP6531 State Management
R6
DNP
R7
22k
EMF Feedback
STM_BEMF_REF
STM_BEMF_A
STM_BEMF_B
STM_BEMF_C
R?
10k
1k
Title
Single motor driver instance; 3 phase BLDC controller.
Size
A4
Number
Revision
1
Date:
8.31.2026
Sheet 2of 4
File:
C:\Users\..\MotorDriverChannel.SchDoc
Drawn By:
Lasse Züngel